MKL Performance Over Time

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Sourcing MKL Versions

- Intel only releases from the last year on their software center.
- Releases through 2023 through Intel's apt repository
- Some 2020 releases on Archive.org
- Official MKL python wheels on pypi include dynamic libraries as far back as 2018
  - All benchmarks used dynamic linking as fewer static libraries could not be sourced

Experimental Setup

Benchmarks

- One warm-up iteration
- 16 timed iterations
- Plots show average time
- Error bars show standard deviation
- Sequential and OpenMP variants
  - 80 threads for OpenMP builds
  - 2025.0 build of Intel's OpenMP runtime used for all benchmarks

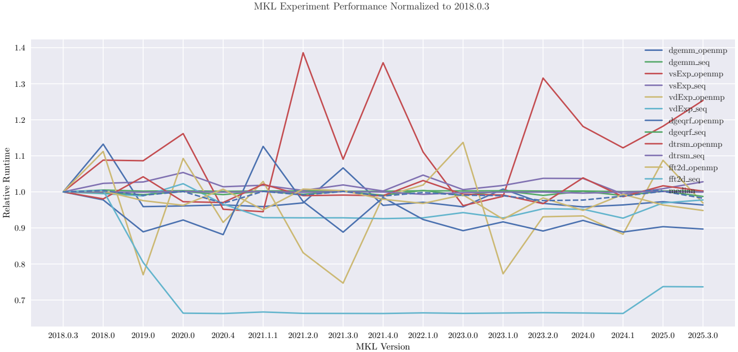
Machine

- IceLake node on TACC
- Intel Xeon Platinum 8380 (Q2'21)
- 80 cores on two sockets

Workloads sizes differ for most experiments between sequential and OpenMP builds

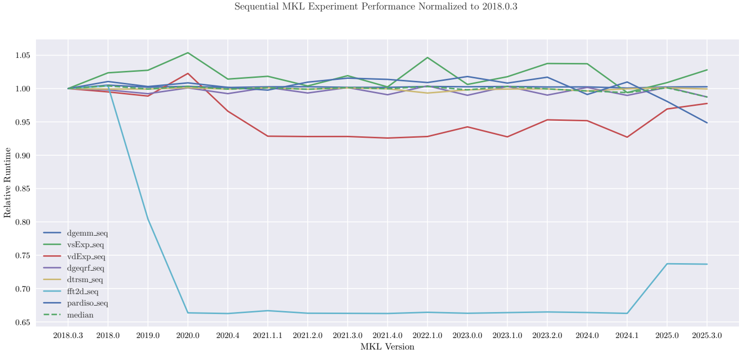
Overview - All

- All experiments (except pardiso - OpenMP)
- Performance normalized to 2018.0.3
- 2025 average is 0.98
- Median normalized performance as dashed line



Overview - Sequential

- All sequential experiments
- Performance normalized to 2018.0.3
- 2025 average is 0.95
- Median normalized performance as dashed line



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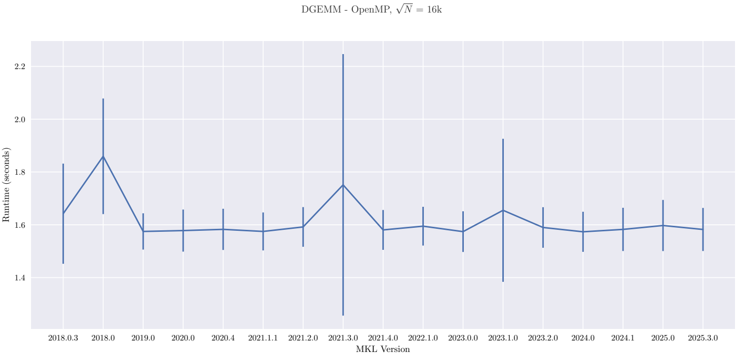
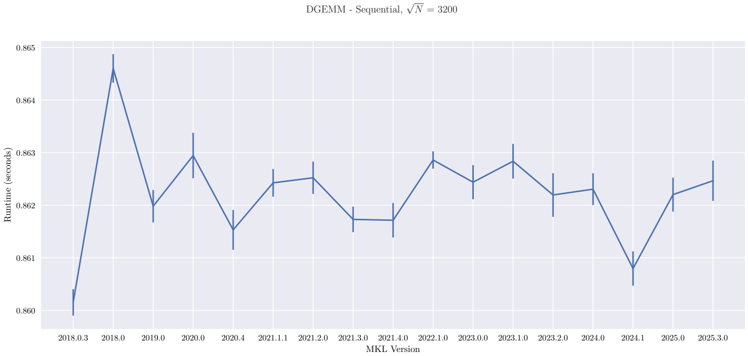
dgemm

- Double precision general matrix multiply
- $C = \alpha A \cdot B + \beta C$
- Less noise in sequential runtimes - expected
- Different Workload sizes,
  - $\sqrt{N} = 16000$ - OpenMP
  - $\sqrt{N} = 3200$ - Sequential

dgemm - OpenMP



dgemm - Sequential



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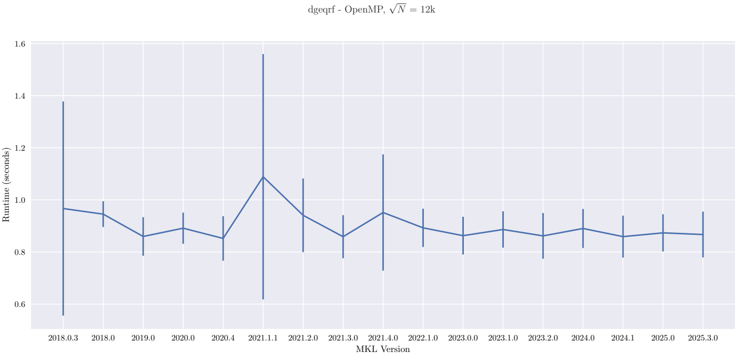
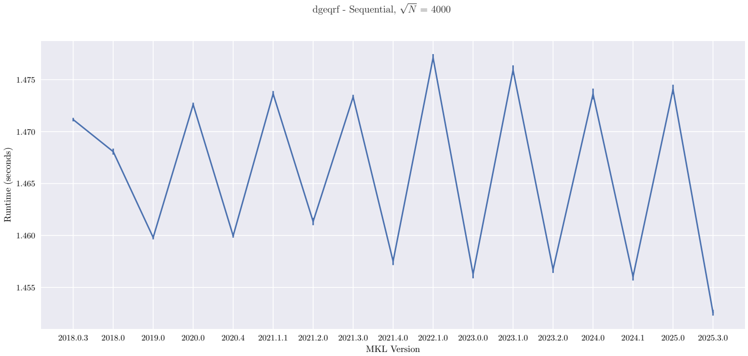
dgeqrf

- Double precision QR factorization
- Different Workload sizes,
  - $\sqrt{N} = 12000$ - OpenMP
  - $\sqrt{N} = 4000$ - Sequential

dgeqrf - OpenMP



dgeqrf - Sequential



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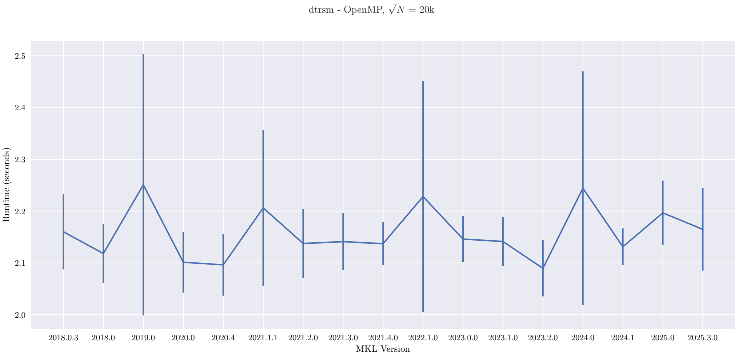
dtrsm

- Double precision solve for

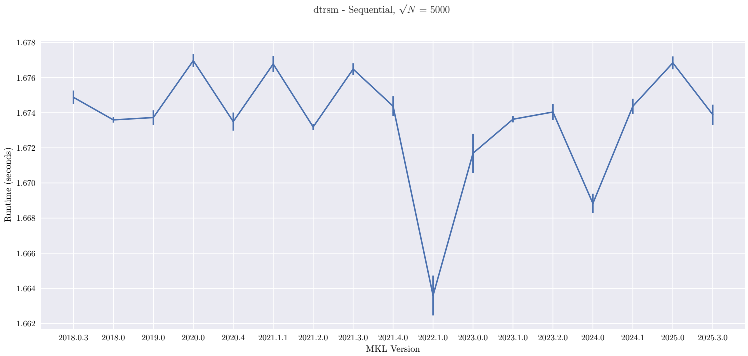
$$\mathrm{op}(A) \cdot X = \alpha \cdot B$$

- Different Workload sizes,
  - $\sqrt{N} = 20000$ - OpenMP
  - $\sqrt{N} = 5000$ - Sequential

dtrsm - OpenMP



dtrsm - Sequential



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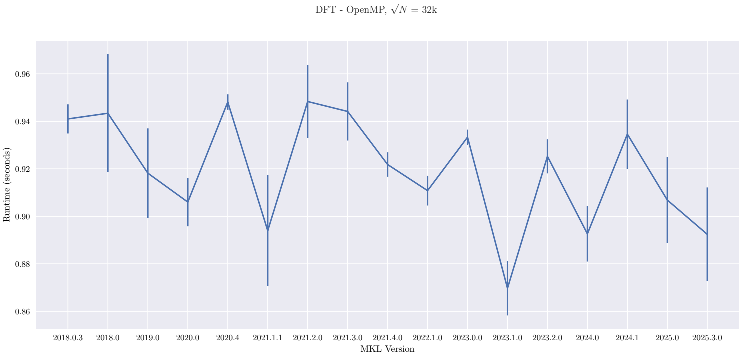
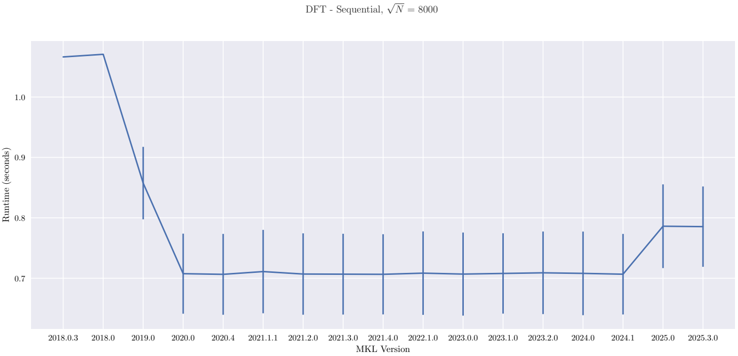
DFT 2d

- Double precision discrete Fourier transform
  - General improvement over time
- Trend is very clear for sequential
- From late 2018 release notes (build not tested)
  - Improved performance for batched real-to-complex 3D for Intel® Xeon® Processor supporting Intel® Advanced Vector Extensions 512 (Intel® AVX-512) (codename Skylake Server)
  - Improved performance with and without scaling factor across all domains.
- Different Workload sizes,
  - $\sqrt{N} = 32000$ - OpenMP
  - $\sqrt{N} = 8000$ - Sequential

DFT 2D - OpenMP



DFT 2D - Sequential



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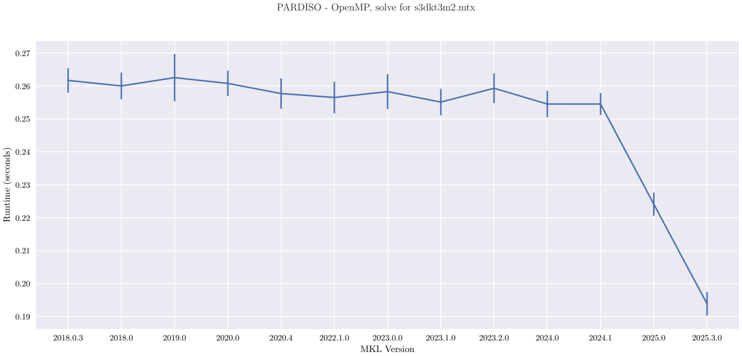
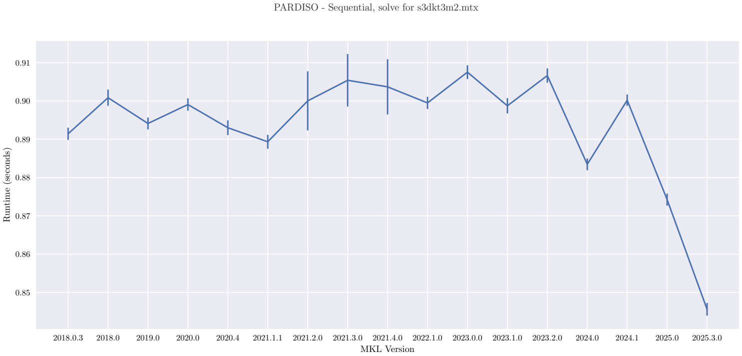
DFT 2d

- Double precision direct sparse linear solver
- Solve `s3dkt3m2.mtx`
- Improvement starting in 2025
- From release notes:
  - "Improved iterative refinement feature of oneMKL PARDISO"
- Same workload for OpenMP and Sequential
  - OpenMP builds for 2021 were unable to solve benchmark

Pardiso - OpenMP



Pardiso - Sequential



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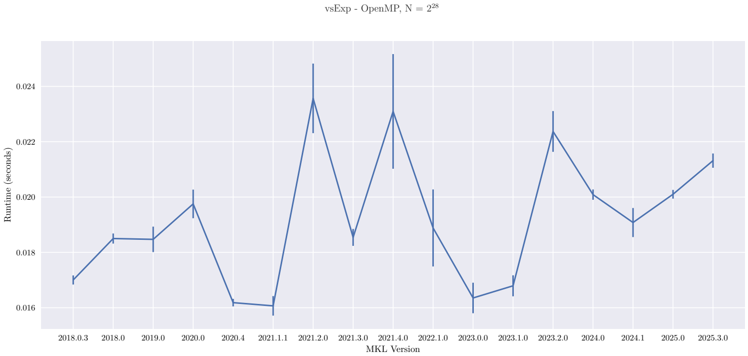
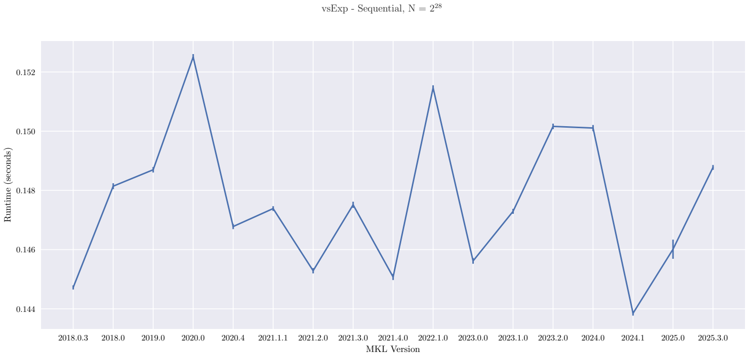
vsExp

- Single precision element-wise exponential
- $a_i = e^{b_i}$
- Performance has degraded over time
- Same workload size for sequential and OpenMP

vsExp - OpenMP



vsExp - Sequential



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vdExp

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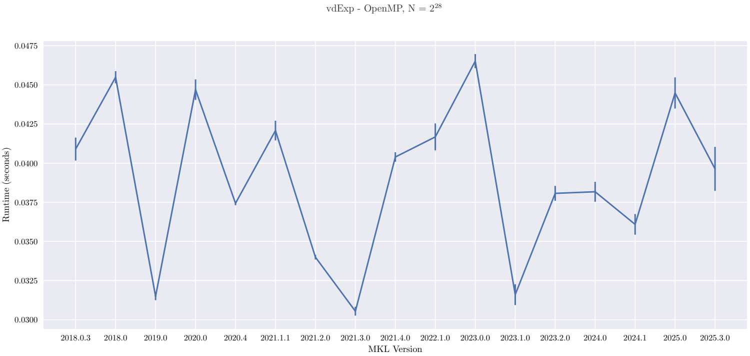
vdExp - OpenMP



vdExp - Sequential